

Estimating Hallucination Risk in Black-Box Large Language Model Answers with Calibrated and Explainable Machine Learning

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Abstract

Large language models answer in fluent text, and some answers are not supported by the evidence that is available. A confidently wrong answer can mislead users in question answering and content review. Current detectors have limits. Judge models are slow and costly. Encoder classifiers need model weights and a GPU. Feature-based systems rarely report calibration or cross-domain results. This study presents a lightweight black-box model that estimates hallucination risk, $P(\text{hallucinated} \mid q, c, a)$, from a question, a reference context, and a candidate answer. The aim is a score that is accurate, calibrated, explainable, and fast without access to model internals. The proposed model extracts 26 evidence-consistency features in seven groups: lexical overlap, entities, bidirectional NLI entailment and contradiction, numbers, hedging, semantic similarity, and length. A tuned XGBoost classifier consumes these features. Platt and isotonic calibrators are fitted on the validation split only. SHAP attributions explain single predictions. Every experiment runs on a grouped 70/15/15 split with seeds 42, 123, and 456. Group-aware splits keep both answer variants of a question in one partition. This removes a leakage path that raises F1 by 0.004. The feature design keeps inference on a CPU and keeps each explanation readable. On HaluEval QA (20,000 rows), the model reaches $F1 = 0.9846$, $AUROC = 0.9979$, and $ECE = 0.0045$. Target recalibration cuts the ECE on natural RAGTruth QA from 0.8185 to 0.1335. Zero-shot transfer still fails ($F1 = 0.3022$), and the paper reports that failure with its length-related cause. One analysis takes 61.8 ms and costs near \$0.001 per 1,000 predictions, about 100 times less than the measured LLM judge. A FastAPI service and a Next.js dashboard deploy the system with claim-level evidence checks.

1. Introduction

Hallucination detection is an active research area. SelfCheckGPT samples multiple responses and measures their consistency without an external database [1]. Luna uses a small evaluation model that costs less than a large judge [2]. Semantic entropy groups sampled answers by meaning and flags confabulation from disagreement, again with no external knowledge [3]. A hybrid framework combines frozen encoder features with lightweight classifiers and reports strong benchmark scores [4]. Two practical problems remain. Judge models are slow and expensive, and encoder models are hard to interpret. Feature-based detectors are cheaper, but their probabilities are rarely calibrated and their behavior outside the training domain is rarely tested. Two recent surveys group these families and list evaluation and reliability limits among the open questions [5, 6]. This study builds a black-box model that reads a question, a reference context, and a candidate answer and estimates $P(\text{hallucinated} \mid q, c, a)$. The model uses 26 evidence-consistency features in seven groups and a tuned XGBoost classifier, with Platt and isotonic calibration and SHAP explanations. HaluEval QA provides the training and in-domain test data. RAGTruth supplies natural responses for the transfer and recalibration checks, and FaithBench adds a summarization stress test.

Six recent studies are closest to this work. HaluAgent shows that small language models can select tools and detect multiple hallucination types, yet each answer needs multiple tool calls [7]. Valentin et al. report a calibrated multi-scoring detector with a clear cost analysis, and the detector still depends on LLM scoring for its main signal [8]. Sundaragiri et al. combine retrieval grounding, NLI verification, and an XGBoost classifier, and their evaluation reports classification metrics only [9]. Haq et al. score hallucination with token-level SHAP-LIME attributions, without testing whether those attributions stay stable [10]. Deng et al. study cross-domain detection and show that detectors lose accuracy outside their training domain, and their method targets multi-turn dialogue signals [11]. FaithBench

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reports that modern LLMs still produce hard summarization hallucinations, and it is a summarization benchmark with no risk-calibration study [12].

Four issues remain open in the literature. Calibration is not reported for lightweight detectors, explanation quality is not measured, cross-domain behavior is not tested, and error patterns are not analyzed case by case. This study addresses the four issues together. Group-aware splits and artifact-control baselines cover the benchmark shortcuts, and validation-only calibration answers the reliability question. Zero-shot runs on RAGTruth and FaithBench measure the transfer limits. HaluEval QA is suitable for training because it provides paired correct and hallucinated answers for 10,000 questions, and RAGTruth provides natural responses with human span annotations for an external check. The accepted outcomes are a calibrated risk model with $F1 = 0.9846$ and $ECE = 0.0045$ on HaluEval QA, a target recalibration that reduces the RAGTruth QA ECE from 0.8185 to 0.1335, a reported zero-shot failure at $F1 = 0.3022$, and a deployed system that completes one analysis in 61.8 ms for near \$0.001 per 1,000 predictions.

2. Conventional Method

2.1. Preliminaries

A hallucination is text that is fluent but not supported by the available evidence, and detection quality is usually judged with labeled benchmark splits, precision, recall, F1, and AUROC. Two evaluation choices matter for deployment. The first is calibration, since a risk score is only useful when its numeric value matches the observed frequency of hallucinated answers, and expected calibration error and reliability diagrams expose the gap between the two [13]. The second is interpretability, since a user who sees a high-risk flag needs to know which evidence signal produced it, and feature-attribution methods such as SHAP provide that view for tabular models [14]. The present work evaluates both properties in a black-box setting, where the detector reads the question, an optional reference context, and the candidate answer, and never inspects the generating model.

2.2. Existing Methods

Hallucination detection is an active research area, and the existing methods fall into three broad families: asking a language model to evaluate output consistency, training neural classifiers over frozen encoders, and building on hand-crafted features and interpretable models. Recent surveys organize the same families in more detail and collect their open problems [5, 6]. LLM-based detection relies on model-internal signals or on a separate judge model. SelfCheckGPT [1] samples multiple responses and treats their agreement as a hallucination signal, and HaluAgent [7] shows that small language models can act as cheaper detectors. Semantic entropy measures uncertainty over the meaning of those samples instead of their surface form [3]. Luna [2] takes the judge idea further with a lightweight evaluation model that matches larger judges at a fraction of the cost, and a calibrated multi-scoring detector reaches competitive accuracy with an explicit cost analysis [8]. None of these judge verdicts gives a feature-level reason, which makes a single failure hard to debug.

A second line of work trains classifiers on top of pretrained encoders, and a hybrid framework in [4] combines encoder features with lightweight models and reports strong results on hallucination benchmarks. These methods are accurate, but they need the encoder weights, they run on GPUs, and their embeddings are not directly interpretable. The third family keeps the pipeline interpretable by construction. A multi-signal framework in [9] combines retrieval grounding, NLI verification, and an XGBoost classifier, and [10] scores hallucination with token-level SHAP-LIME attributions. Small verifier models such as MiniCheck reach high fact-checking accuracy at low cost, which shows that verification does not always need a large judge [15]. The present work differs from [10] in an important way, since it explains predictions at the level of semantically meaningful evidence-consistency features rather than raw tokens, and it adds calibrated risk scores and a cross-domain evaluation that both papers lack.

On the benchmark side, HaluEval [16] and TruthfulQA [17] are standard test beds, FActScore [18] measures factual precision in long-form generation, and RAGTruth [19] contributes naturally generated responses with human word-level annotations that are closer to production data. FaithBench [12] shows that current detectors still struggle on challenging modern-LLM outputs, and SpikeScore [11] demonstrates that cross-domain generalization remains an open problem.

2.3. Research Gap

Prior work has explored hallucination detection, feature engineering, calibration, and SHAP in isolation. Three gaps remain open. First, token-level attributions in [10] explain raw text rather than semantically meaningful evidence

signals, and the attributions are visualized without any reliability measurement. Second, calibration is rarely reported for lightweight detectors, and target-domain behavior is rarely tested, while the closest multi-signal framework [9] reports classification metrics only. Third, the benchmark itself imposes limits, since around 85% of HaluEval is generated with a sampling-then-filtering procedure [20], and such data blurs the line between factually wrong and hallucinated content. Cross-domain evaluation remains open as well, and SpikeScore [11] shows that detectors trained in one domain lose accuracy in another. This study fills these gaps by testing calibrated, explainable risk prediction under domain shift with a reproducible artifact.

3. Proposed Method

Figure 1 shows the full proposed pipeline at a glance. The system takes a question, an optional reference context, and a candidate answer produced by a black-box LLM, and it returns a calibrated risk probability together with per-feature explanations. Each stage is described in the following subsections.

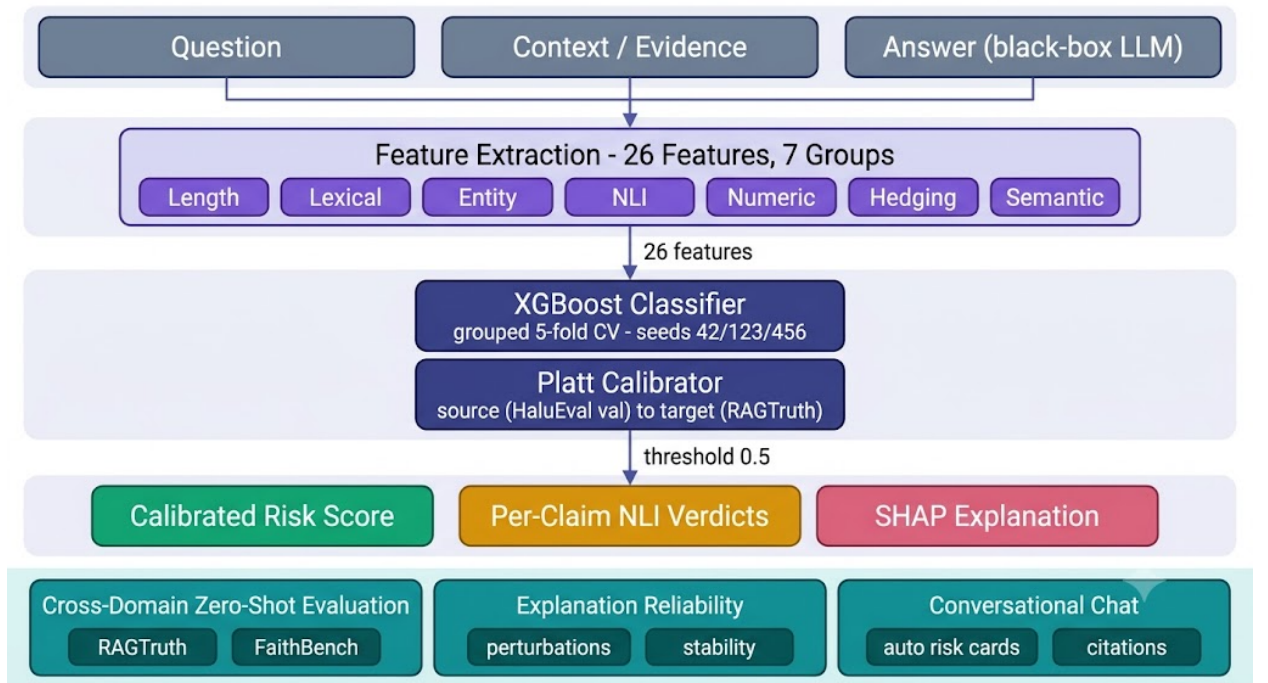


Figure 1: End-to-end pipeline of the proposed model. Evidence-consistency features from the question, context, and answer feed a grouped-CV trained, Platt-calibrated XGBoost model. Outputs include a calibrated risk score, per-claim NLI verdicts, and SHAP explanations, with cross-domain and explanation-reliability evaluation.

3.1. Task Definition

Given a question q , an optional reference context c (evidence), and a candidate answer a produced by a black-box LLM, the task is to predict the probability $P(\text{hallucinated} \mid q, c, a)$. The proposed model predicts risk, not truth. It flags answers that are unsupported by, or contradictory to, the available evidence.

3.2. Features

Table 1 lists the 26 features in seven groups. NLI signals come from a DeBERTa-v3 cross-encoder [21] trained on SNLI [22] and MultiNLI [23], evaluated in both directions, context-to-answer and answer-to-context, which yields six probability features. Semantic similarity uses MiniLM sentence embeddings [24, 25], entity features use spaCy NER [26], and the numeric and hedging groups are collected with regular expressions and a small hedging lexicon.

Table 1

Feature groups (26 features total).

Group	Features	Source
Length/style	n_chars, n_words, n_sentences, avg_word_len	regex
Lexical	overlap vs. context & question, Jaccard $\times 2$	token sets
Entity	entity counts, overlap ratio, novel-entity ratio	spaCy NER
NLI	entail/contradict/neutral $\times 2$ directions	cross-encoder NLI
Numeric	number counts, overlap ratio, novel numbers	regex
Hedging	hedge count, hedge density	lexicon
Semantic	cosine(context,answer), cosine(question,answer)	MiniLM

3.3. Models

Eight models are compared. The overlap heuristic sets risk to one minus the lexical overlap between the answer and the context, and its threshold is tuned on validation. Logistic regression on standardized features and a random forest with 300 trees are the two standard classifiers. The proposed classifier is XGBoost [27], trained on the same features. Three controls isolate the value of the feature design, an NLI-only logistic regression that sees only the six NLI probabilities, an answer-only TF-IDF linear model, and a TF-IDF linear model over the full input. A majority baseline that predicts the negative class for every row confirms the balance of the labels. XGBoost hyperparameters come from a randomized search of 30 iterations over depth, learning rate, tree count, subsample, and column subsampling, scored with 5-fold stratified cross-validation on the training split. The best configuration is depth 4, learning rate 0.01, 500 trees, subsample 0.9, and column subsampling 0.7, with a cross-validated AUROC of 0.9970. Every experiment is repeated with seeds 42, 123, and 456, and results are reported as the mean over seeds.

3.4. Calibration

Calibrators are fit on the validation split only. Platt scaling fits a sigmoid to the raw scores [28], and isotonic regression fits a step function [29], and both are compared against the raw model on the test split with expected calibration error over 10 bins [13], Brier score, and reliability diagrams. A second step targets a new domain. When a small labeled sample of the target corpus is available, the calibrator is re-fit on that sample, and the effect is measured on a disjoint held-out part of the target corpus.

3.5. Explainability and Reliability

Global interpretability comes from SHAP [14]. A TreeExplainer over the raw XGBoost model produces mean absolute SHAP importance and beeswarm summaries, and local explanations show the feature contributions behind a single verdict. Explanation reliability is measured directly in this work. Each test answer is perturbed in six ways, by replacing entities, numbers, or dates, removing supporting context, inserting irrelevant context, and shuffling clauses, and the calibrated score should stay stable under neutral perturbations and react to semantic ones. Stability is quantified with Spearman and Kendall rank correlation between the original and perturbed scores, the fraction of top-1 flips, and the top- k Jaccard overlap of the SHAP feature rankings.

3.6. System Implementation

The methods above are deployed as a two-tier web application, and Figure 2 shows the architecture. The Next.js frontend streams an LLM conversation on the chat page using the assistant-ui library, and the FastAPI backend holds the model, the calibrator, the SHAP explainer, and the feature models, all loaded once at startup. Every machine-learning request from the frontend goes through a backend-for-frontend proxy in the Next.js layer to the FastAPI endpoints, which avoids cross-origin issues and keeps the API key on the server side.

The chat page uses the streaming route of the Vercel AI SDK. A tool call from the conversation model triggers the analysis endpoint, and the result renders as a risk card inside the message thread. The card shows the calibrated score

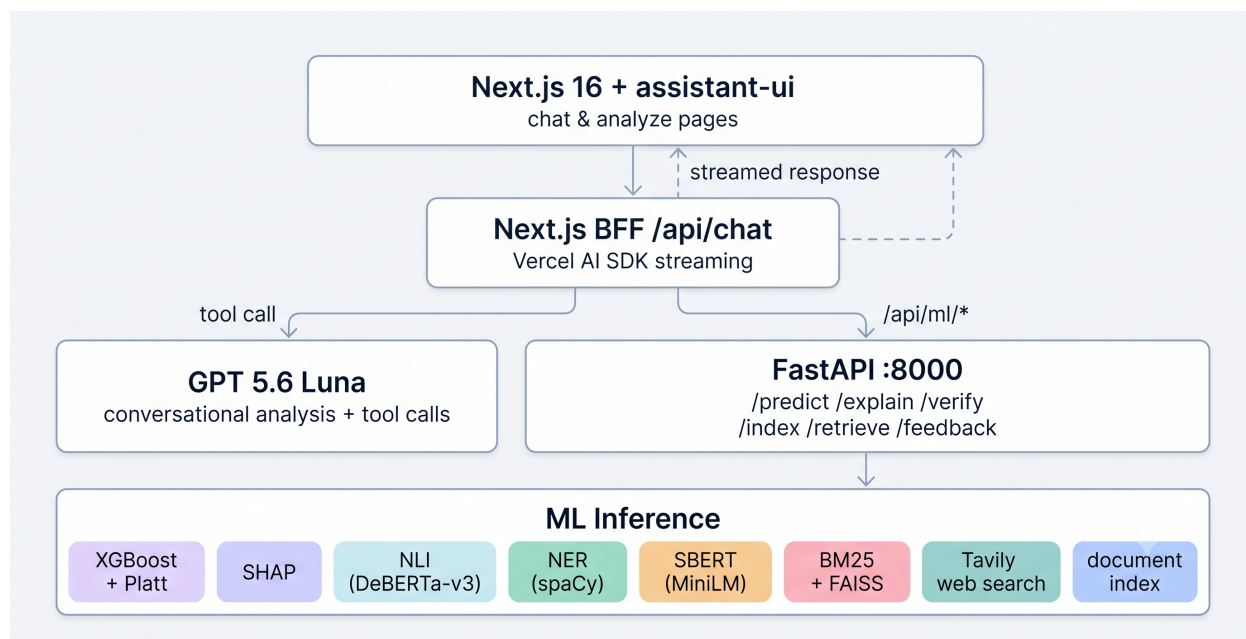


Figure 2: System architecture of the proposed model. The Next.js backend-for-frontend proxies machine-learning requests to FastAPI and streams LLM chat with tool-generated UI widgets.

with its threshold band, the per-claim verdicts, the SHAP contribution bars, and a link to a full evidence panel that expands the supporting and contradicting passages.

Beyond the score, the system verifies the answer claim by claim. Each claim is decomposed from the answer text and checked with sentence-level NLI against one of three evidence sources, the provided context, an indexed document collection, or live web search. The verdicts supported, contradicted, and unsupported are rendered per claim, with the driving evidence sentence quoted for contradicted claims. Web search uses the Tavily API, and the document index combines BM25, FAISS dense retrieval, and reciprocal rank fusion with a cross-encoder reranker. When the verification score falls into an uncertain band, the claim is routed to an LLM judge that returns a verdict with a short reasoning note.

The verdict headline leads the card. Any contradicted claim sets the risk to high, an unsupported claim sets it to medium, and fully supported claims set it to low, while the XGBoost score appears as the legacy model score beneath it with a notice when the two disagree. This ordering follows directly from the style-sensitivity result in Section 5, where a full-sentence grounded answer can saturate the calibrated score even though every claim is supported. The displayed percentage itself comes from the evidence-domain calibrator, which is fitted on natural RAGTruth responses rather than on HaluEval, and in chat it is adjusted by the claim verdicts, so grounded full-sentence answers score low while answers that contradict the evidence score high.

The analyze page accepts typed text or an uploaded file, runs the pipeline, and shows the risk gauge, the calibrated score, the SHAP bars with the seven feature groups expandable, and a comparison against the raw model score. A demo page walks through the whole flow without an API key. The experiment dashboard renders the tables of Section 5 directly from the artifacts folder, and an about page reads the frozen manifest to show the model version, the feature version, and the training commit. Feedback on the verdicts is stored in a feedback log, and a threshold-tuning script turns that log into updated verification thresholds.

Figure 3 shows two analysis cases from the analyze page, a hallucinated number that the raw model scores at 0.999 and the evidence-domain calibrator reports at 0.47, and a grounded answer that scores below 0.01, together with the SHAP contribution charts from the raw model.

3.7. Reproducibility

The source repository contains the complete pipeline. `src/data` holds the download and split scripts, `src/features` the seven feature groups, `src/models` the training protocol together with the cross-domain,

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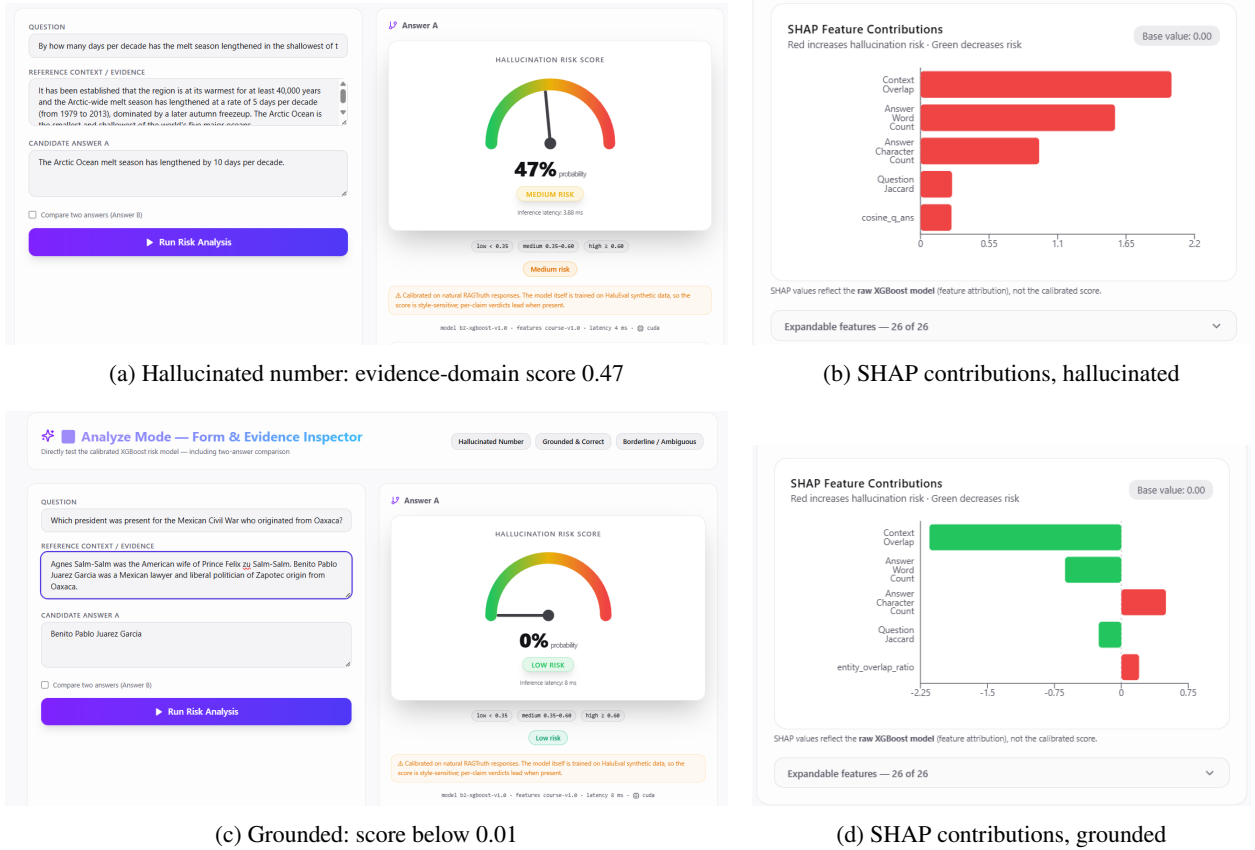


Figure 3: Analyze page cases with the raw-model SHAP contribution charts. Top: a hallucinated number, where the context states 5 days per decade and the answer states 10. Bottom: a grounded answer.

calibration-shift, and explanation-reliability runs, `src/explain` the SHAP analysis, and `src/api` the FastAPI application. `requirements.txt` pins every dependency with exact versions. The train, validation, and test split indices are saved to `artifacts/split_indices.json` and `.npz`, all seeds are centralized in the run configuration, and the model, calibrator, scaler, and SHAP explainer artifacts are stored under `artifacts/models/`.

The full experiment protocol is reproducible end to end. A self-contained Colab notebook (`colab/HaluRISC-Training_Version_B.ipynb`) embeds the runtime source, verifies file hashes before execution, runs the pipeline on a GPU, and persists the artifacts to Google Drive. The `run_all_experiments.py` script reproduces the B-run locally from a frozen configuration (`configs/version_b.yaml`), `make_manifest.py` freezes the artifact hashes into `manifest.frozen.json`, and `verify_artifacts.py` checks that every reported number matches the frozen artifacts. The manuscript itself is built from the same files that the dashboard reads, so the tables in Section 5 trace back to the artifact files, including the length-binned error table and the reviewed audit sheet.

The environment is Python 3.12 on Windows 11 with an NVIDIA RTX 3060 6 GB GPU (CUDA 12.8, torch 2.11.0+cu128) and 32 GB of RAM, using scikit-learn 1.9.0, XGBoost 3.3.0, SHAP 0.52.0, spaCy 3.8.14, and sentence-transformers 5.6.1. The frontend uses Next.js 16.3.0 with pnpm.

4. Experiment

4.1. Dataset

The QA subset of HaluEval [16] is the training source. It contains 10,000 questions, each with a knowledge passage, a correct answer, and a hallucinated answer, which gives 20,000 labeled rows. The rows are split 70/15/15 at the question-group level, producing 14,000 training, 3,000 validation, and 3,000 test rows. This grouped, leakage-free

design matters because both answer variants of a question share the same context, and a random split could otherwise let the model memorize the pairing. The split indices are saved to disk for exact reproducibility.

Transfer is measured on RAGTruth [19], a corpus of naturally generated responses with human word-level span annotations, and on FaithBench [12], a summarization hallucination benchmark built from modern LLM outputs. The RAGTruth QA task contributes 5,934 rows in 989 groups, of which 5,034 rows (839 groups) form the recalibration set and 900 rows (150 groups) the held-out test, with no overlapping groups between the two. Table 2 summarizes the corpora.

Table 2

Datasets used for training and evaluation.

Corpus	Rows	Groups	Use
HaluEval QA	20,000	10,000	train / validation / test
RAGTruth QA	5,934	989	zero-shot + recalibration
RAGTruth full	17,790	2,965	zero-shot
FaithBench	750	750	zero-shot

Text is whitespace-normalized. Lexical features operate on lowercased token sets, while NER and NLI keep the original casing. Missing context is handled explicitly, with lexical overlap set to zero and NLI probabilities to one third. All three corpora are used under their original licenses: HaluEval and RAGTruth are MIT-licensed, and FaithBench is distributed under CC BY-NC-SA 4.0. Raw files are not redistributed. The repository ships the download scripts, source revisions, and SHA-256 hashes instead, so every split can be regenerated from the original repositories.

4.2. Evaluation Protocol

Models are compared on precision, recall, F1, AUROC, PR-AUC, and MCC, and calibration with expected calibration error over 10 bins, Brier score, and negative log-likelihood. Hyperparameters are chosen by a randomized search of 30 iterations with grouped 5-fold cross-validation on the training split. Every experiment is repeated with seeds 42, 123, and 456 and reported as the mean over seeds. Significance is assessed with McNemar’s test on paired test predictions, bootstrap 95% confidence intervals from 1,000 resamples, and a Wilcoxon signed-rank test across seeds. The group structure is respected everywhere, so the two rows of one question never appear in both the training and the evaluation folds.

4.3. Calibration Protocol

Calibrators are fit on the validation split only, and the comparison of raw, Platt-scaled, and isotonic probabilities happens on the test split. The target-domain protocol follows the same principle. The recalibration set of 5,034 RAGTruth QA rows is used only to fit the target calibrator, and all reported numbers come from the disjoint 900-row test set.

4.4. Explainability Protocol

Global explanations come from a SHAP TreeExplainer [14] over the raw XGBoost model. Three measurements check whether the explanations are trustworthy. First, the mean absolute SHAP ranking is compared against permutation importance with a Kendall tau correlation. Second, the top-5 feature set is re-estimated from 1,000 bootstrap resamples and compared with the Jaccard index. Third, two perturbation studies are run on the test answers. Content edits replace entities, numbers, or dates, remove the supporting context, or insert an irrelevant sentence, and each edit should move the calibrated score if it changes the evidence. Structure edits shuffle the clause order, and these should leave the score nearly unchanged. For each operator the mean absolute score change, the fraction of large score changes, the top-1 SHAP flip rate, and the Spearman correlation between the original and perturbed scores are recorded. A neutralization study sets the top- k SHAP features of each sample to its median and tracks the mean score change for $k = 1, 3, 5, 10$.

4.5. Efficiency and Judge Protocol

Latency is measured on 200 test samples on the reference laptop with an RTX 3060 6 GB GPU, where NLI and embeddings run on the GPU and the remaining features on the CPU. The LLM-as-judge comparison uses GPT 5.6 Luna on the same 200 samples with a fixed prompt, and the cost is computed from the recorded token counts.

5. Results

5.1. Model comparison

Table 3 reports the test-set performance of every baseline, averaged over the three seeds. XGBoost reaches an F1 of 0.9846 with an AUROC of 0.9979, ahead of the random forest by 0.0004 in F1 and the logistic regression by 0.0186. The overlap heuristic, which needs no training beyond its threshold, still reaches an F1 of 0.9352, and a majority baseline that predicts the negative class for everything scores zero, which confirms that the labels are balanced. The NLI-only baseline lags at 0.6714, which shows that the lexical, length, and semantic signals carry information the NLI cross-encoder alone does not. The TF-IDF variants sit between these extremes, with an answer-only linear model at 0.9224 and a full-input model at 0.5999. The margin between XGBoost and the random forest is small but significant in a McNemar test on the paired test predictions ($p = 0.044$), while the comparison against logistic regression ($p = 0.43$) and the heuristic ($p = 0.35$) is not significant. The Wilcoxon test across the three seeds finds no difference, which is expected given that the test compares only three points. Bootstrap confidence intervals for XGBoost are F1 [0.9812, 0.9895] and AUROC [0.9964, 0.9989].

The split design is verified against a known confound. A row-level split, where the two rows of one question can fall into different partitions, inflates F1 to 0.9886, and the grouped split corrects this to 0.9846, a drop of 0.004 in F1 and 0.0001 in AUROC. The reported numbers all use the grouped split, so the margin is small but real.

Table 3

Test-set performance on HaluEval QA, mean over seeds 42/123/456 ($n = 3,000$). The ECE and Brier columns measure the reliability of each model's risk score.

Model	P	R	F1	AUROC	PR-AUC	MCC	ECE	Brier
Heuristic (1-overlap)	0.947	0.923	0.935	0.913	0.820	0.872	0.354	0.267
Logistic regression	0.977	0.955	0.966	0.993	0.990	0.933	0.010	0.025
Random forest	0.990	0.979	0.984	0.998	0.998	0.969	0.014	0.013
NLI-only	0.648	0.697	0.671	0.718	0.753	0.319	0.068	0.218
TF-IDF (answer-only)	0.955	0.892	0.922	0.970	0.967	0.852	0.083	0.066
TF-IDF (full input)	0.640	0.565	0.600	0.675	0.679	0.248	0.068	0.233
XGBoost (ours)	0.993	0.976	0.985	0.998	0.998	0.970	0.004	0.012

The comparison so far uses aggregate metrics. Table 4 shows how the same evidence produces different risk scores at the level of single answers. Cases 1 and 2 share one question and one context. Case 1 is a grounded answer span, and case 2 is a hallucinated sentence that reuses words from the context. The overlap heuristic separates the pair poorly (0.0000 against 0.0769), because the hallucinated sentence copies context words, and the NLI-only model fails in the same way (0.3928 against 0.3950). The answer-only TF-IDF model flags the hallucinated case at 0.9121 but also puts the grounded span at 0.7490. Logistic regression, random forest, and XGBoost separate the pair, with 0.4975 against 0.9990, 0.5593 against 0.9924, and 0.4541 against 0.9996. Case 3 is a grounded one-word answer. The heuristic scores it at 1.0000, logistic regression at 0.7109, the random forest at 0.7634, and the raw XGBoost at 0.9135, so four models flag a correct answer as risky. The deployed calibrator maps these raw probabilities to 0.0000 and removes the false alarm, which is why the system displays the calibrated score and keeps the raw model score as a secondary signal.

Table 4

Risk scores for three HaluEval test cases, seed 42. Cases 1 and 2 share one question and one context: case 1 is a grounded answer span, and case 2 is a hallucinated sentence that reuses context words. Case 3 is a grounded one-word answer. The last row is the deployed calibrated score. A low score is correct for cases 1 and 3, and a high score is correct for case 2. Values are read from `b2_case_scores.csv`.

Model	Risk score		
	Case 1	Case 2	Case 3
Overlap heuristic	0.0000	0.0769	1.0000
Logistic regression	0.4975	0.9990	0.7109
Random forest	0.5593	0.9924	0.7634
NLI-only	0.3928	0.3950	0.3058
TF-IDF (answer-only)	0.7490	0.9121	0.1694
XGBoost (raw)	0.4541	0.9996	0.9135
Deployed (calibrated)	0.0000	0.4265	0.0000

5.2. Calibration on the source domain

Table 5 compares raw, Platt-scaled, and isotonic probabilities on the HaluEval test set. The raw model is already well calibrated, with an ECE of 0.0045, and neither recalibration method improves it. Platt scaling increases the ECE to 0.0091 and isotonic regression to 0.0070, and both raise the negative log-likelihood. This is an honest negative result. XGBoost with a logistic objective produces reliable probabilities on this benchmark, so the calibration stage adds no value in the source domain, and the reliability diagrams in Figure 4 confirm that the raw curve tracks the diagonal closely.

Table 5

Calibration on the HaluEval test set (mean over seeds).

Method	F1	ECE	Brier	NLL
Raw XGBoost	0.985	0.0045	0.0117	0.045
Platt (sigmoid)	0.985	0.0091	0.0129	0.058
Isotonic	0.984	0.0070	0.0124	0.073

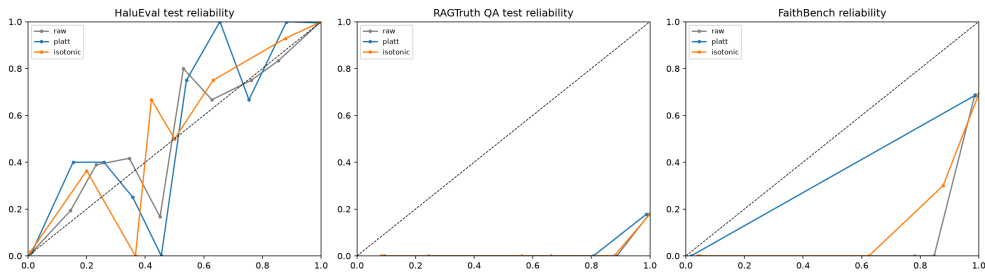


Figure 4: Reliability diagrams of the raw, Platt-calibrated, and isotonic models on the HaluEval test set (left), RAGTruth QA (middle), and FaithBench (right).

5.3. Target-domain recalibration

The calibration story changes completely on RAGTruth. Table 6 applies the model to the 900-row RAGTruth QA test set and compares the raw scores, the source calibrator fitted on the HaluEval validation split, and calibrators re-fitted on 5,034 RAGTruth rows. The raw and source-calibrated probabilities are badly off, with an ECE above 0.8, because almost every sample is predicted with near-certainty. Re-fitting the calibrator on the target data cuts the ECE to 0.1335 (Platt) and 0.1316 (isotonic) and the Brier score from 0.801 to 0.164. Figure 5 shows the shift. One deployment caveat is visible in the same numbers. After target recalibration the predicted positive rate drops to nearly zero at the fixed 0.5 threshold, so a deployment must also retune the decision threshold. The improvement is real for the probabilities, and the labels follow only after that second step.

Table 6

Target-domain recalibration on RAGTruth QA (calibration $n = 5,034$, test $n = 900$).

Calibrator	ECE	Brier	NLL	Pred. pos.
Raw scores	0.8185	0.8164	5.20	0.999
Source Platt (HaluEval)	0.8089	0.8011	3.62	0.999
Target Platt (RAGTruth)	0.1335	0.1639	0.51	0.000
Target isotonic	0.1316	0.1663	0.53	0.000

5.4. Cross-domain zero-shot transfer

Table 7 reports the zero-shot evaluation with no training or adaptation on the target corpora. The recall of 1.0 on every row tells the main story. The model flags almost everything as hallucinated, the precision lands on the label prior, and the AUROC stays near 0.5. This is the central negative result of the paper. Synthetic HaluEval hallucinations

Calibrated Hallucination Risk Estimation



Figure 5: Target-domain recalibration on RAGTruth QA. Left and middle: raw and target-calibrated score distributions on the 900-row test set, with the 0.5 threshold marked. Right: ECE and Brier before and after recalibration.

differ systematically from naturally generated responses, and the RAGTruth numbers quantify that gap. FaithBench sits between the two extremes, with an F1 of 0.8130 at an AUROC of 0.5314, since its summaries are shorter and closer to the HaluEval format. The RAGTruth task breakdown shows the same pattern, with F1 of 0.4515 on QA, 0.4603 on summarization, and 0.8140 on data-to-text, where answers mirror the structured input closely. Context length also matters. The F1 rises from 0.4940 on responses with under 128 characters of context to 0.7091 on responses with 128 to 511 characters, which suggests the model needs enough evidence text to work with. Figure 6 plots the score distributions behind the table.

Table 7

Zero-shot transfer, mean over seeds (recall is 1.0 on every corpus).

Corpus	n	F1	AUROC	PR-AUC	MCC	ECE
RAGTruth QA	900	0.302	0.540	0.190	0.016	0.819
RAGTruth all	17,790	0.603	0.497	0.407	0.041	0.560
FaithBench	750	0.813	0.531	0.683	0.107	0.301

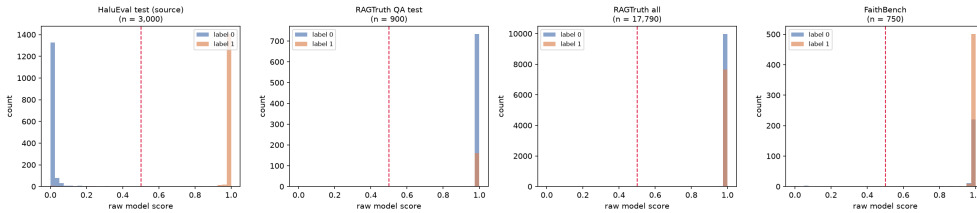


Figure 6: Raw model score distributions on the HaluEval test set and the zero-shot corpora, with the 0.5 threshold marked. The source panel separates the classes, while every zero-shot panel piles both labels at 1.0.

5.5. Ablation

Table 8 removes each feature group and retrains from scratch with the same protocol. Lexical features are the dominant group, with the F1 dropping by 0.0247 when they are removed. Length contributes a smaller but consistent amount, NLI a modest one, and the entity, numeric, and hedging groups are neutral, with deltas within 0.0002 of zero. Removing the semantic group slightly improves the F1, a negative result reported rather than hidden. The ranking agrees with the SHAP analysis in the next subsection.

5.6. Explanation reliability

Figure 7 shows the mean absolute SHAP values for the top-12 features. The context overlap dominates, followed by the answer length features, the context-to-answer contradiction probability, and the question-to-answer cosine. The explanation set agrees with the ablation in spirit, since lexical and length features carry the largest weights. Three checks assess whether the explanations are stable. The SHAP ranking agrees with permutation importance at a Kendall tau of

Table 8

Feature-group ablation, mean over seeds (full model F1 = 0.9846, AUROC = 0.9979).

Removed group	F1	Δ F1	AUROC
length	0.9801	−0.0045	0.9963
lexical	0.9600	−0.0247	0.9923
entity	0.9847	+0.0001	0.9982
nli	0.9823	−0.0023	0.9973
numeric	0.9844	−0.0002	0.9982
hedging	0.9848	+0.0001	0.9981
semantic	0.9860	+0.0014	0.9982

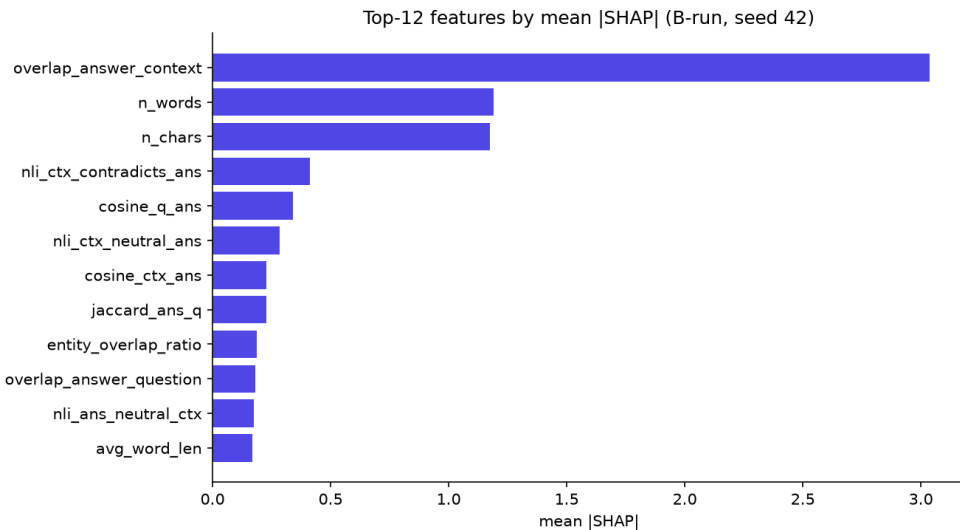
0.66. The top-5 feature set is identical across 1,000 bootstrap resamples, with a Jaccard of 1.0. Neutralizing the top- k features drops the mean score monotonically from -0.079 at $k = 1$ to -0.412 at $k = 10$, which confirms that the top features are actually driving the predictions rather than decorating them.

Table 9 reports the perturbation study. Content edits move the score in proportion to the semantic weight of the edit. Replacing an entity changes the score by 0.360 on average, a number by 0.345, a date by 0.279, and removing the supporting context by 0.272. The irrelevant insert and clause shuffle leave the score nearly untouched, at 0.006 and 0.001. The top-1 SHAP feature never flips under any operator, and the Spearman correlation stays between 0.84 and 0.95. The sample sizes are small for dates and clause shuffles, 13 and 14 respectively, because not every answer contains an edit-able element, and the table reports n openly.

Table 9

Perturbation stability of the calibrated score and SHAP ranking.

Operator	n	Δ score	Large rate	Top-1 flip	Spearman
Entity replacement	78	0.360	0.372	0.000	0.840
Number replacement	22	0.345	0.364	0.000	0.852
Date replacement	13	0.279	0.308	0.000	0.870
Support removal	66	0.272	0.288	0.000	0.874
Irrelevant insert	120	0.006	0.008	0.000	0.948
Clause shuffle	14	0.001	0.000	0.000	0.921

**Figure 7:** Mean absolute SHAP values for the top-12 features (B-run, seed 42).

A structured expert audit tested these explanations case by case. The B5 run exported 26 error-enriched cases, namely 9 false positives, 10 false negatives, and 7 cases inside a calibrated band from 0.35 to 0.65, and two review passes were performed with an AI assistant, which the released sheet records in a review-mode field. Both passes judged the explanation against the answer-context evidence rather than the dataset label. The passes agreed on 23 of 26 cases (88%), and both marked 15 cases implausible, with the remaining cases split between plausible (6 and 7) and unsure (5 and 4). The three disagreements all sit on borderline scores. The audit adds a qualitative reading to the numbers above. False positives inherit the length confound of the benchmark, and false negatives occur when a wrong answer reuses entities that appear in the context, which keeps lexical overlap high. Two audited cases (items 2251 and 2436) carry a hallucinated label even though the context directly supports the answer, so they look like benchmark label noise rather than model errors. The audit sample is deliberately error-heavy, so these proportions describe the audit and not the test set.

5.7. Efficiency and cost

Table 10 breaks down the latency over 200 test samples. The NLI cross-encoder and the spaCy entity pass dominate, at 27.1 ms and 21.1 ms median, while the model itself needs 1.3 ms and SHAP 1.9 ms. The total is 61.8 ms at the median, and the deployed model artifact weighs 0.82 MB. The marginal cost of a local prediction is about \$0.001 per 1,000 calls, since the only consumable resource is electricity.

Table 10

Per-component latency in ms ($n = 200$).

Component	p50	p95	mean
Core lexical	0.10	0.15	0.11
Entity (spaCy NER)	21.13	47.24	25.34
NLI (2 directions)	27.09	68.60	40.09
Semantic (MiniLM)	9.25	15.55	12.56
XGBoost inference	1.26	8.80	2.62
SHAP explanation	1.91	15.52	3.65
Total per sample	61.77		

5.8. LLM-as-judge comparison

Table 11 compares the deployed model with GPT 5.6 Luna as a judge on the same 200 samples. The judge reaches an F1 of 0.841, the XGBoost model 0.985, and the two agree on 84.5% of the cases. The paired McNemar test is highly significant ($p = 1.6 \times 10^{-5}$), and the discordant pairs are strongly one-sided, with 28 cases where the judge is wrong and the model is right against 3 where the roles are reversed. The cost gap is the more practical difference. The judge costs \$0.105 per 1,000 predictions and needs 1.3 seconds at the median, while the model costs about \$0.001 per 1,000 and runs in 61.8 ms.

Table 11

LLM-as-judge vs. XGBoost on 200 test samples.

Model	Acc	P	R	F1	p50 (ms)	Cost/1K
GPT 5.6 Luna judge	0.860	0.974	0.740	0.841	1310	\$0.105
XGBoost (ours)	0.985	0.980	0.990	0.985	62	≈\$0.001

5.9. Error analysis

The 3,000-sample test set contains 48 errors, 12 false positives and 36 false negatives, an error rate of 1.6%. A manual inspection of 20 sampled errors assigns them to two categories. Short-answer ambiguity covers very short answers where neither the lexical nor the NLI evidence is decisive, and label ambiguity covers answers that are semantically plausible under both labels. Table 12 reports the counts. Both categories trace back to the construction of the benchmark, where short answers are frequently hallucinated by design.

The length distribution behind those errors is the more informative signal. Table 13 bins the full test set by answer length. The hallucinated share is 2.1% among one-word answers and rises to 99.2% above 17 words, so answer length

Table 12

Error taxonomy for 20 sampled misclassifications.

Category	FP	FN
Label ambiguity	7	4
Short-answer ambiguity	3	6

is close to a perfect proxy for the label in HaluEval. The model reproduces the pattern: the mean score climbs from 0.024 to 0.993 across the bins, two-thirds of all false positives fall in the 5-8 word bin at a 10.6% false-positive rate, and false negatives concentrate in the 1-4 word bins at about an 18% rate. This mechanism explains the zero-shot transfer result. Natural RAGTruth responses are full sentences, so the model reads their length as evidence of hallucination, which is why recall reaches 1.0 and AUROC stays near 0.5 on that corpus. The deployed interface responds to this failure mode by leading with per-claim evidence verdicts and presenting the model score as a secondary signal.

Table 13

Test-set errors by answer length, mean over seeds (17+ contains only two negative rows).

Words	Rows	Hallucinated	FP rate	FN rate	Mean score
1	524	0.021	0.004	0.182	0.024
2	514	0.033	0.000	0.176	0.035
3-4	474	0.127	0.002	0.178	0.126
5-8	641	0.902	0.106	0.031	0.888
9-16	600	0.982	0.000	0.003	0.980
17+	247	0.992	0.333	0.000	0.993

5.10. Discussion and limitations

The results need to be read against multiple limitations. Around 85% of HaluEval is generated with a sampling-then-filtering procedure [20], so its hallucination patterns are synthetic by construction, and the zero-shot transfer result on RAGTruth shows how far in-distribution performance is from natural responses. Length and overlap statistics are highly informative on HaluEval but they are benchmark artifacts that may not carry over to other settings, a warning that the ablation study and the SHAP analysis both reinforce. The benchmark also reduces hallucination to a binary label, which leaves graded severity and partially supported answers out of scope. The audit in Section 5 found two cases where the hallucinated label contradicts the context, so a small amount of label noise remains in the benchmark. Every feature in the pipeline is English-centric because the NER, NLI, and embedding models are trained on English data, and the supervised model itself is trained on a single dataset, so domain adaptation remains open work. Calibration deserves one more honest note. On the source benchmark the raw probabilities are already well calibrated, and neither Platt nor isotonic regression improves them, while the target-domain recalibration on RAGTruth cuts the ECE there from 0.8185 to 0.1335. The deployed artifact ships the Platt-calibrated model to follow the project default. Finally, the calibrated score is style-sensitive in conversational use, since full-sentence answers can saturate it even when the per-claim evidence verdicts are supported. The interface therefore leads with the evidence verdicts and presents the model score as a secondary signal. Two established directions point to the next step. Automated evaluation suites for retrieval-augmented answers, such as RAGAs, score claim support and answer relevance without human review [30], and small verifier models make claim-level checks cheap enough to run on every answer [15]. Both directions fit the verification layer of the deployed system.

5.11. Ethical considerations

A hallucination-risk score predicts risk, not truth, and it must not be treated as a fact-checker. Over-reliance on automated detectors creates false trust, especially when scores are confidently wrong out of domain, which is exactly what the RAGTruth result demonstrates. Detection quality varies across domains and languages, and the current system is English-only. The interface therefore labels scores as risk estimates and shows a training-data warning with every prediction.

6. Conclusion

Hallucination detection needs to be cheap, fast, and explainable before it is useful in practice. This study presented a lightweight black-box risk predictor that combines 26 engineered evidence-consistency features with a tuned XGBoost classifier, calibrated probabilities, and SHAP explanations. On HaluEval QA it reaches an F1 of 0.9846 and an AUROC of 0.9979 at an expected calibration error of 0.0045, runs in 61.8 ms per analysis at the median, and costs about \$0.001 per 1,000 predictions, roughly 100 times less than a GPT 5.6 Luna judge that it outperforms on the same 200 samples. A target-domain recalibration on 5,034 RAGTruth rows cuts the ECE on that corpus from 0.8185 to 0.1335, which shows that the calibration stage pays off exactly where the source-domain result said it would not.

The negative results are reported with the same care as the positive ones. The raw model needs no recalibration on HaluEval, the entity and hedging features add no measurable value, and zero-shot transfer to natural RAGTruth responses fails, with an F1 of 0.3022 on the QA split. The paper therefore frames the supervised risk model as a source-domain estimator, and the deployed system adds claim-level verification with evidence retrieval to cover the transfer gap. Future work targets multicalibration under distribution shift, cross-domain evaluation on FaithBench [12], and neural baselines for comparison.

Declarations

- **Conflicts of Interest:** The authors declare no conflicts of interest.
- **Data availability:** The raw evaluation corpora are not redistributed. The repository ships the download scripts, source revisions, SHA-256 hashes, and license notes for HaluEval (MIT), RAGTruth (MIT), and FaithBench (CC BY-NC-SA 4.0), so every split and feature matrix can be regenerated from the original sources. Model artifacts, split indices, and the frozen experiment manifest are included with the code release.
- **Code availability:** The training pipeline, FastAPI inference service, and Next.js dashboard are available in the project repository.

CRedit authorship contribution statement

Md. Atikur Rahaman: Conceptualization, Methodology, Software, Writing - original draft. **Saiful Alam Sabbir:** Data curation, Investigation, Validation, Visualization. **MD. Miraz Ahamed:** Software, Formal analysis, Writing - review & editing. **Ohidujjaman Tuhin:** Supervision, Validation, Writing - review & editing.

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